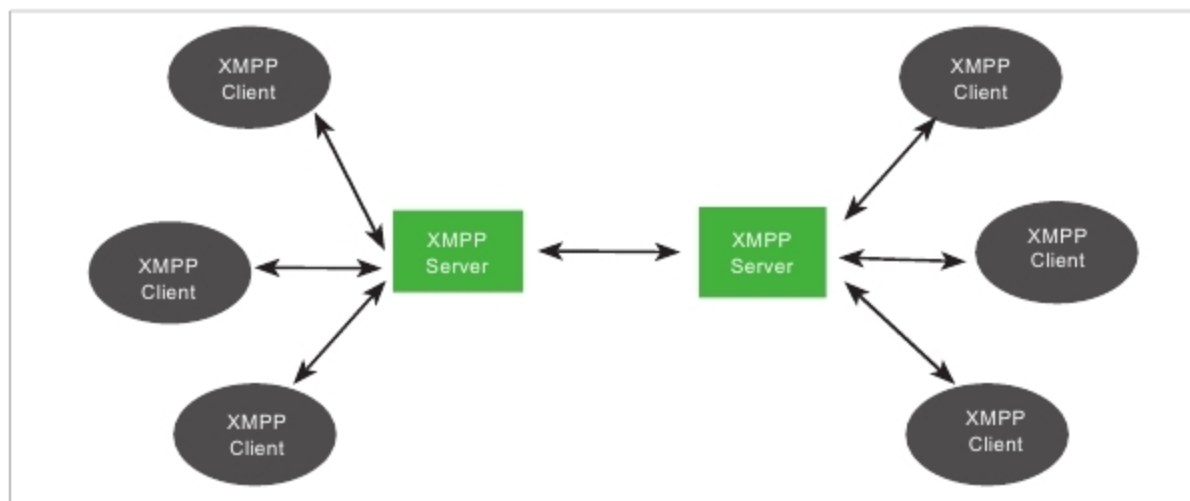
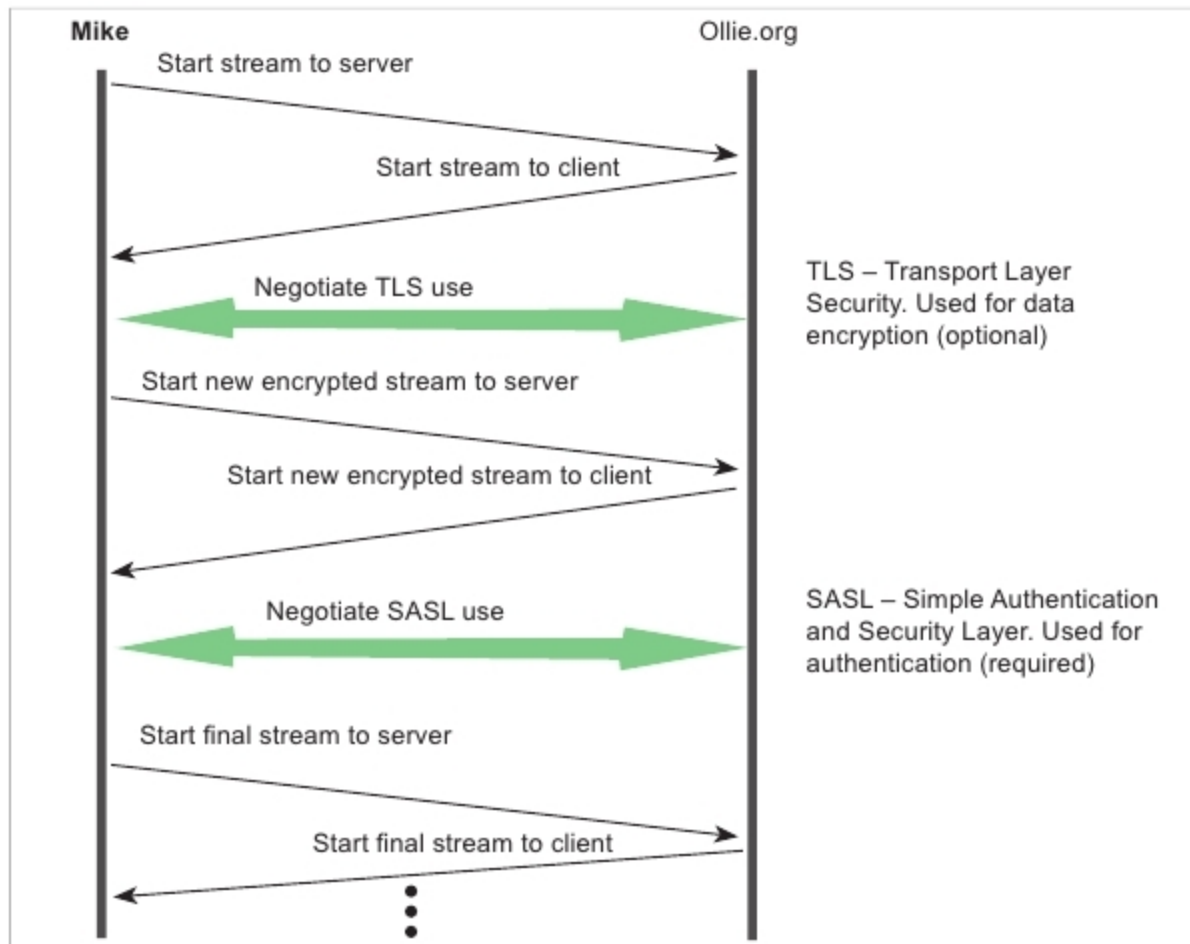




manufacturers connected to different platforms can talk to each other with a standard communication protocol.

- MQTT has different levels of quality of service. This flexibility is not available in XMPP.
- MQTT deployments become difficult to manage when the number of devices increases, while XMPP scales very easily.

The pros and cons of XMPP

Pros

- Addressing scheme to recognise devices on the network
- Client-server architecture
- Decentralised

- Flexible
- Open standards and formalised

Cons

- Text-based messaging and no provision for end-to-end encryption
- No provision for quality of service
- The data flow is usually more than 70 per cent of the XMPP protocol server, of which nearly 60 per cent is repeated; the XMPP protocol has a large overhead of data to multiple recipients
- Absence of binary data

- Limited scope for stability

How the XMPP protocol manages communication between an XMPP client and server

The features of the XMPP protocol that impact communication between the XMPP client and the XMPP server are described in Figure 1.

Figure 2 depicts an XML message swap between client 'Mike' and server 'Ollie.org'.

- XMPP uses Port 5222 for the client to server (C2S) communication.
- It utilises Port 5269 for server to server (S2S) communication.
- Discovery and XML streams are used for S2S and C2S communication.
- XMPP uses security mechanisms such as TLS (Transport Layer Security) and SASL (Simple Authentication and Security Layer).
- There are no in-between servers for federation, unlike e-mail.

Direct messaging is used as a method for immediate message transmission to and reception from online users (Figure 3).

XMPP via HTTP

As an alternative to the TCP protocol, XMPP can be used with HTTP in two ways: polling and binding. The polling method, now deprecated, essentially implies that messages stored on a server-side database are acquired by an XMPP client by way of HTTP 'GET' and 'POST' requests. Binding methods are applied using bi-directional streams over synchronous HTTP permit servers to push messages to clients as soon as they are sent. This push model of notification is more efficient than polling, wherein many of the polls return no new data.

XMPP provides a lot of support for communication, making it well suited for use within the realm of the Internet of Things. **END** 🐧

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